APEX Autopilot

Institutional-grade, paper-only autonomous execution for cross-market prediction arbitrage (Kalshi × Polymarket) with a fail-fast risk stack (M01–M09), multi-agent intelligence, and a Bloomberg-style operator terminal.

Repository: github.com/aaravjj2/Autopilot-public

Hackathon / judge materials: **HACKATHON.md** · Pitch brief: **docs/pitch/APEX_Autopilot_Pitch_Brief_and_Speech.md**

What it does

APEX Autopilot runs a disciplined **L0–L4 pipeline**:

Layer	Role
L0	Ingestion — Kalshi, Polymarket, retries, cache hydration
L1	Finance brain — scoring, spreads, confidence
L2	Agent panel — multi-agent thesis / verdicts
L3	Execution + risk — dual-leg paper routing, **M01 paper-first** gates
L4	Observability — audit stream, telemetry

The **Next.js terminal** (`autopilot-local/frontend`) provides Arb Radar, Autopilot proposals, risk views, and live audit feeds.

Quickstart (full stack)

Requirements: Python 3.11+ with `uvicorn`, Node 18+, `npm`

```
git clone https://github.com/aaravjj2/Autopilot-public.git
cd Autopilot-public
python -m venv .venv && source .venv/bin/activate   # optional but recommended
pip install -e ".[dev]"
cp keys.env.example keys.env    # API keys (gitignored)
cp .env.example .env            # feature flags

# Start API (:8000), marketplace (:8001), frontend (:3000), scheduler
bash start_all.sh
```

Service	URL
Terminal	http://localhost:3000
APEX API docs	http://127.0.0.1:8000/docs
Health	http://127.0.0.1:8000/health
Arb opportunities	http://127.0.0.1:8000/api/arb/opportunities

Stop everything:

```
bash stop_all.sh
```

Verify APEX owns port 8000

Another local app may already use `:8000`. APEX health includes `proposals` and `timestamp` — not a generic `"service"` string from unrelated APIs.

```
curl -s http://127.0.0.1:8000/health | python3 -m json.tool | head -20
curl -s http://127.0.0.1:8000/api/arb/opportunities | python3 -c "import sys,json;d=json.load(sys.stdin
```

If health looks wrong or arb returns 404:

```
bash scripts/ensure_apex_port_8000.sh
bash start_all.sh
```

Judge demo (no live venue keys)

```
export DEMO_MODE=true
python scripts/seed_demo.py
PYTHONPATH=src python -m uvicorn backend_api:app --host 127.0.0.1 --port 8000
# Other terminal:
cd autopilot-local/frontend && npm install && npm run dev
```

Open **http://localhost:3000/dashboard/arb-radar**

Submission / demo artifacts

Record screenshots + walkthrough video for Devpost:

```
bash start_all.sh
bash scripts/build_submission_artifacts.sh
```

Outputs under `artifacts/submission/`:

- `APEX_Autopilot_Demo.mp4` — product tour
- `screenshots/` — gallery images
- `api-*.json` — live API proof
- `pitch/` — PDF, PPTX, spoken brief
- `APEX_Autopilot_Submission_Artifacts.zip` — one-click bundle (generated at repo root)

Tests

```
python -m pytest tests/ -v
cd autopilot-local/frontend && npx playwright test
```

Roadmap day-file verifier:

```
python scripts/verification/verify_roadmap_daily.py --start 1 --end 20
```

Configuration

- Secrets: `keys.env` (gitignored) — see `keys.env.example`
- Settings: `.env` / Pydantic `src/apex/core/config.py`
- Integrations under `external/` — external/README.md

Paper trading only — `M01_PAPER_REQUIRED` runs first on every arb execution path.

Strict integration mode:

```
export STRICT_INTEGRATIONS=true
apex-engine
```

Architecture & roadmap

- Deep architecture: docs/architecture/master_plan_arch.md
- Execution control plane: master_plan.md
- Daily runbooks: `one-year-daily/day-001.md` … `day-260.md`

Other entry points

Command	Purpose
`apex-engine`	Core engine CLI
`apex-scheduler`	Background loops
`apex-healthz`	Health on `:8088`
`apex-dashboard`	Streamlit ops `:8501`
`bash scripts/start-apex-stack.sh`	Copy-trading stack variant

Details: autopilot-local/README.md · AGENTS.md

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